

Java AWT / Swing Programming Assignment

ALNAJEEB CSA0911 192424325

Q1. Find Square (25 marks)

Design an Applet that accepts a number from the user and displays its square when a button is clicked. Use AWT controls and appropriate event handling.

Source Code (FindSquare.java)

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

/*
<applet code="FindSquare.class" width="350" height="150"></applet>
*/
public class FindSquare extends Applet implements ActionListener {
    Label l1, l2;
    TextField tf1, tf2;
    Button b1;

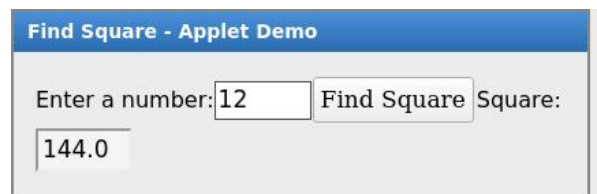
    public void init() {
        setLayout(new FlowLayout());
        l1 = new Label("Enter a number:");
        tf1 = new TextField(10);
        b1 = new Button("Find Square");
        l2 = new Label("Square: ");
        tf2 = new TextField(10);
        tf2.setEditable(false);

        add(l1); add(tf1); add(b1); add(l2); add(tf2);
        b1.addActionListener(this);
    }

    public void actionPerformed(ActionEvent ae) {
        try {
            double num = Double.parseDouble(tf1.getText());
            double sq = num * num;
            tf2.setText(String.valueOf(sq));
        } catch (NumberFormatException e) {
            tf2.setText("Invalid input");
        }
    }

    // Allow running as a standalone application (java.applet is legacy/removed
    // in newer JDKs for browser use, so we host it in a Frame for testing)
    public static void main(String[] args) {
        Frame f = new Frame("Find Square - Applet Demo");
        FindSquare app = new FindSquare();
        app.init();
        f.add(app);
        f.setSize(400, 150);
        f.addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) { System.exit(0); }
        });
        f.setVisible(true);
    }
}
```

Sample Output



Q2. Menu Driven Calculator (1 marks)

Create a menu-driven Swing application for a simple calculator (Add, Subtract, Multiply, Divide, Clear, Exit) using MenuBar, Menu, and MenuItem.

Source Code (Calculator.java)

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

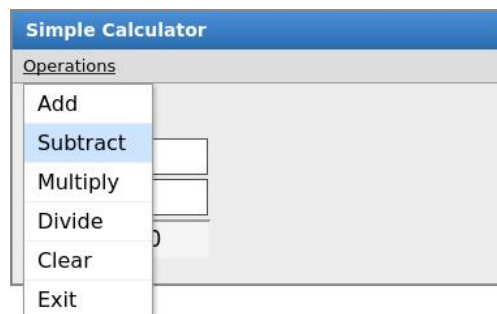
public class Calculator extends JFrame implements ActionListener {
    JTextField tf1, tf2, tfResult;

    public Calculator() {
        setTitle("Simple Calculator");
        setLayout(new FlowLayout());
        JMenuBar menuBar = new JMenuBar();
        JMenu menu = new JMenu("Operations");
        String[] items = {"Add", "Subtract", "Multiply", "Divide", "Clear", "Exit"};
        for (String item : items) {
            JMenuItem mi = new JMenuItem(item);
            mi.addActionListener(this);
            menu.add(mi);
        }
        menuBar.add(menu);
        setJMenuBar(menuBar);
        tf1 = new JTextField(8); tf2 = new JTextField(8);
        tfResult = new JTextField(8); tfResult.setEditable(false);
        add(new JLabel("Num1:")); add(tf1);
        add(new JLabel("Num2:")); add(tf2);
        add(new JLabel("Result:")); add(tfResult);
        setSize(350, 150);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        String cmd = e.getActionCommand();
        try {
            if (cmd.equals("Exit")) { System.exit(0); }
            if (cmd.equals("Clear")) { tf1.setText(""); tf2.setText(""); tfResult.setText(""); return; }
            double a = Double.parseDouble(tf1.getText());
            double b = Double.parseDouble(tf2.getText());
            double res = 0;
            switch (cmd) {
                case "Add": res = a + b; break;
                case "Subtract": res = a - b; break;
                case "Multiply": res = a * b; break;
                case "Divide":
                    if (b == 0) { tfResult.setText("Error: /0"); return; }
                    res = a / b; break;
            }
            tfResult.setText(String.valueOf(res));
        } catch (NumberFormatException ex) {
            tfResult.setText("Invalid input");
        }
    }

    public static void main(String[] args) { new Calculator(); }
}
```

Sample Output



Q3. Swing Application - Layout Managers (25 marks)

Develop a Swing application that demonstrates the use of the following layout managers: FlowLayout, BorderLayout, and GridLayout. Switch among layouts using radio buttons.

Source Code (LayoutDemo.java)

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class LayoutDemo extends JFrame implements ActionListener {
    JPanel panel;
    JRadioButton rbFlow, rbBorder, rbGrid;

    public LayoutDemo() {
        setTitle("Layout Manager Demo");
        setLayout(new BorderLayout());
        JPanel radioPanel = new JPanel();
        rbFlow = new JRadioButton("FlowLayout", true);
        rbBorder = new JRadioButton("BorderLayout");
        rbGrid = new JRadioButton("GridLayout");
        ButtonGroup bg = new ButtonGroup();
        bg.add(rbFlow); bg.add(rbBorder); bg.add(rbGrid);
        radioPanel.add(rbFlow); radioPanel.add(rbBorder); radioPanel.add(rbGrid);
        rbFlow.addActionListener(this);
        rbBorder.addActionListener(this);
        rbGrid.addActionListener(this);
        add(radioPanel, BorderLayout.NORTH);
        panel = new JPanel();
        panel.setLayout(new FlowLayout());
        addButtons();
        add(panel, BorderLayout.CENTER);
        setSize(400, 200);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setVisible(true);
    }

    void addButtons() {
        for (int i = 1; i <= 5; i++) panel.add(new JButton("Button " + i));
    }

    public void actionPerformed(ActionEvent e) {
        panel.removeAll();
        if (rbFlow.isSelected()) panel.setLayout(new FlowLayout());
        else if (rbBorder.isSelected()) panel.setLayout(new BorderLayout());
        else panel.setLayout(new GridLayout(2, 3));
        if (rbBorder.isSelected()) {
            panel.add(new JButton("North"), BorderLayout.NORTH);
            panel.add(new JButton("South"), BorderLayout.SOUTH);
            panel.add(new JButton("East"), BorderLayout.EAST);
            panel.add(new JButton("West"), BorderLayout.WEST);
            panel.add(new JButton("Center"), BorderLayout.CENTER);
        } else addButtons();
        panel.revalidate(); panel.repaint();
    }

    public static void main(String[] args) { new LayoutDemo(); }
}
```

Sample Output



Q4. Sources of Events (25 marks)

Write a program that demonstrates different sources of events (Button click, TextField input, Checkbox selection, Window closing). Display the event source and type.

Source Code (EventSourcesDemo.java)

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class EventSourcesDemo extends JFrame
    implements ActionListener, ItemListener, WindowListener {
    JButton button; JTextField textField; JCheckBox checkBox; JTextArea display;

    public EventSourcesDemo() {
        setTitle("Event Sources Demo");
        setLayout(new BorderLayout());
        JPanel top = new JPanel();
        button = new JButton("Click Me");
        textField = new JTextField(12);
        checkBox = new JCheckBox("Subscribe");
        top.add(button); top.add(textField); top.add(checkBox);
        add(top, BorderLayout.NORTH);
        display = new JTextArea(8, 40);
        display.setEditable(false);
        add(new JScrollPane(display), BorderLayout.CENTER);
        button.addActionListener(this);
        textField.addActionListener(this);
        checkBox.addItemListener(this);
        addWindowListener(this);
        setSize(480, 260);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setVisible(true);
    }

    void log(String src, String type) {
        display.append("Event Source: " + src + "\n");
        display.append("Event Type : " + type + "\n\n");
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == button)
            log(e.getSource().getClass().getName() + "[Click Me]", "ActionEvent (Button Click)");
        else if (e.getSource() == textField)
            log(e.getSource().getClass().getName(), "ActionEvent (Enter pressed: \"" + textField.getText() + "\")");
    }

    public void itemStateChanged(ItemEvent e) {
        String st = (e.getStateChange() == ItemEvent.SELECTED) ? "SELECTED" : "DESELECTED";
        log(e.getSource().getClass().getName() + "[Subscribe]", "ItemEvent (Checkbox " + st + ")");
    }

    public void windowClosing(WindowEvent e) {
        log("JFrame", "WindowEvent (Window Closing)");
        System.exit(0);
    }

    public void windowOpened(WindowEvent e) {}
    public void windowClosed(WindowEvent e) {}
    public void windowIconified(WindowEvent e) {}
    public void windowDeiconified(WindowEvent e) {}
    public void windowActivated(WindowEvent e) {}
    public void windowDeactivated(WindowEvent e) {}

    public static void main(String[] args) { new EventSourcesDemo(); }
}
```

Sample Output

